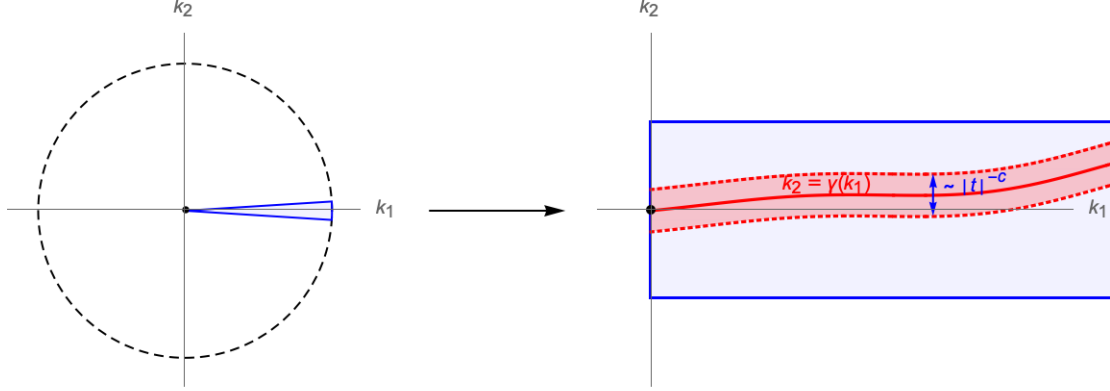


FIGURE 5.1. Frequency domain for $\mathbb{C}_{\mathbf{k} \approx \mathbf{K}_*}(t; v, \theta)$

For the proof, we recall that changing the variable, the sectorial domain of integration for $\mathbb{C}_{\mathbf{k} \approx \mathbf{K}_*}(t; v, \theta)$ is transformed into the rectangular one (see Figure 5.1). To refine the bound in the previous section, we find the curve $k_2 = \gamma(k_1)$ where the modified phase function $\tilde{\varphi}(\mathbf{k}; v, \theta)$ is stationary in the k_2 -direction at the frequency $\mathbf{k} = (k_1, \gamma(k_1))$. Then, near the curve (the red region), we only consider the oscillation in the k_1 -direction, but the additional decay is obtained from the small measure. On the other hand, away from the curve (the blue region), we take the additional decay by integration by parts in the k_2 -direction. The $O(|t|^{-c})$ -width of the red region will be chosen later to optimize the sum of the two bounds.

For this, using Lemma 5.11, we differentiate the phase in the k_2 -direction and write

$$\partial_{k_2} \tilde{\varphi}(\mathbf{k}; v, \theta) = \frac{\sqrt{3}k_1}{2} G(\mathbf{k}), \quad (5.12)$$

where

$$\begin{aligned} G(\mathbf{k}) := & k_2 + \left(-\frac{3b_\theta}{2} - \frac{7a_\theta}{2}k_2 + \frac{15b_\theta}{4}k_2^2 \right) k_1 \\ & + \left(\frac{3a_\theta b_\theta}{4} + \frac{3(44a_\theta^2 - 3)}{32}k_2 - \frac{87}{8}a_\theta b_\theta k_2^2 \right) k_1^2 + k_1^3 \mathcal{O}_0(\mathbf{k}) + k_2^3 \tilde{\mathcal{O}}_0(\mathbf{k}). \end{aligned} \quad (5.13)$$

By the implicit function theorem, we determine the frequencies where $G(\mathbf{k})$ vanishes, which is equivalent to $\partial_{k_2} \tilde{\varphi}(\mathbf{k}; v, \theta) = 0$ when $k_1 > 0$.

Lemma 5.14. *There exist small $\delta_0 > 0$ and an analytic curve $\gamma : (-\delta_0, \delta_0) \rightarrow \mathbb{R}$ such that if $|k_1| < \delta_0$, then $G(k_1, \gamma(k_1)) = 0$ and*

$$\gamma(k_1) = \frac{3b_\theta}{2}k_1 + \frac{9a_\theta b_\theta}{2}k_1^2 + \mathcal{O}_3(k_1),$$

where $\mathcal{O}_3(k_1)$ is an analytic function of k_1 near the origin such that $|\mathcal{O}_3(k_1)| \lesssim |k_1|^3$ (see (2.16)).